

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459253.6704 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.054
Transit depth (Rp/Rs)^2	3.44 +/- 1.99 [%]
Orbital Inclination (inc)	88.47 +/- 1.73
Airmass coefficient 1 (a1)	0.855 +/- 0.061
Airmass coefficient 2 (a2)	0.074 +/- 0.046
Scatter in the residuals of the lightcurve fit is	7.74 %
Best Comparison Star	#2 - [514, 463]
Optimal Aperture	2.53
Optimal Annulus	16.0
Transit Duration (day)	0.1332 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.054 vs 0.1489 (deviation 0.52σ)
Duration fit vs published	0.1332 d vs 0.1304 d (deviation 0.34σ)
Mid-time O–C	7.3 min
Depth SNR	2.7
Residual scatter	7.74 %

Light curve

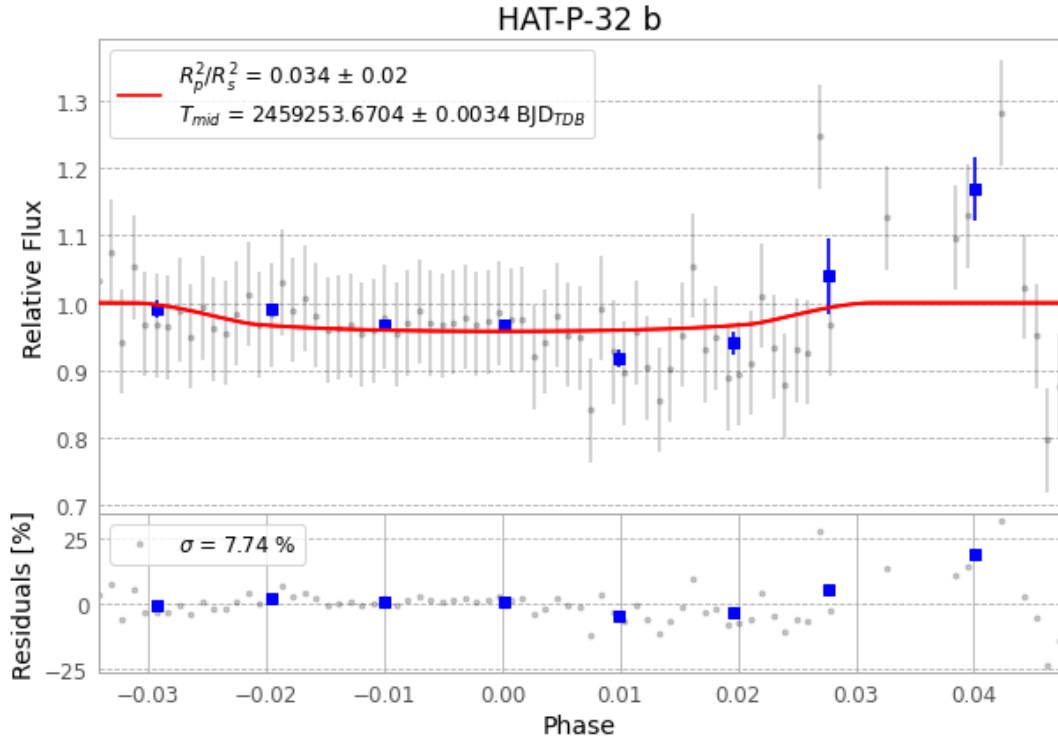


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-02-07.png (SHA-256 22cd05209fe73a07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 328], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 04528cafc8e17072...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-32210208021812.FITS → HATP-32210208063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-210208.log (3378c1f85162...), FinalLightCurve_HAT-P-32 b_2021-02-07.png (22cd05209fe7...), FinalLightCurve_HAT-P-32 b_2021-02-07.csv (5a0f11d8bc10...)

Compute resources

Wall time 546.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:31Z.